Find and Fix the Mistake
Answer Key

Grade 3 — Data Displays and Shapes

Quick Answers

1. (a) the label that belongs there = 4, —
2. (a) the correct number of students = 4, —
3. (a) square corners on that shape = 0, —
4. (a) the correct number of students = 12, —
5. (a) the correct difference = 4, —
6. (a) pairs of parallel sides on that shape = 1, —
7. (a) number of equal gaps = 4, —
8. (a) how many marks are missing = 1, —

What is verified

0 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 — Data Displays and Shapes
3.G.A – *problems 3, 6*
3.MD – *problems 1, 2, 4, 5, 7, 8*
Difficulty 2–5 of 5 across 16 tagged checks.

Common wrong answers

1. *If they got 5: left the uneven scale alone and kept the label the student wrote*
2. *If they got 5: counted the tick marks along the scale instead of the marks stacked above 3*
3. *If they got 4: assumed that four equal sides must mean four square corners*
4. *If they got 15: added the numbers printed along the scale instead of counting the marks*
5. *If they got 6: added the greatest and least values instead of subtracting them*
6. *If they got 2: assumed every four-sided shape has two pairs of parallel sides*
7. *If they got 5: counted the tick marks instead of the gaps between them*

Line Plot P: 1 book has 2 marks, 2 books has 3, 3 books has 4, 4 books has 2, 5 books has 1 — twelve students in all.
Shape Set Q: W is a square, X a rhombus, Y a trapezoid, Z a rectangle.

1. A scale labelled 1, 2, 3, 5, 6. (a) Which number belongs where the 5 is? (b) Why does an uneven scale mislead?

Solution (a): The scale starts at 1 and steps up by 1 each time, so the labels must run $1, 2, 3, 4, 5$. The fourth tick is $1 + 3 \times 1$. $\boxed{4}$

Solution (b) — instructor-judged. A full-credit answer says the steps on a scale have to be the same size, and explains the consequence: if one step is bigger, the marks sit in the wrong places and the picture no longer shows the data fairly. Model answer: "Every step should go up by the same amount. If it jumps from 3 to 5, one gap stands for two books and the others for one, so the plot lies about where the data is." Only saying "the 5 is wrong" is not full credit.

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2. "5 students borrowed 3 books." (a) The correct number. (b) Name the mistake.

Solution (a): Count the marks stacked above 3 on Line Plot P: there are four. $\boxed{4}$

Solution (b) — instructor-judged. The student counted the five numbers printed along the scale instead of the marks above the number 3. A full-credit answer names that swap. Model answer: "They counted the numbers on the bottom line, but those are how many books, not how many students. The students are the $\times$ marks above the number." Restating the right answer is not full credit.

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3. "Shape X is a square because it has four equal sides." (a) Square corners on X. (b) Name the mistake.

Solution (a): Shape X is a rhombus: four equal sides, but its corners are two sharp and two blunt. Testing each with a card corner, none of them fits. $\boxed{0}$

Solution (b) — instructor-judged. A full-credit answer says four equal sides is only half the test — a square needs four square corners as well, and this shape has none, so equal sides alone do not make it a square. Model answer: "It has four equal sides like a square, but a square also needs four square corners. This one is pushed over, so its corners are not square corners. It is a rhombus." Saying only that it looks tilted, with no mention of corners, is not full credit.

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4. Added the scale numbers and got 15. (a) How many students are shown? (b) Name the mistake.

Solution (a): Every student contributed exactly one mark, so count all the marks: $2 + 3 + 4 + 2 + 1 = 12$. $\boxed{12}$

Solution (b) — instructor-judged. The student added $1+2+3+4+5$, which are the numbers of books, not the students. A full-credit answer says what those numbers actually mean and what to count instead. Model answer: "The numbers on the bottom say how many books a student borrowed. To find how many students there are, count the marks, not the labels."

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5. Added greatest and least to get 6. (a) The correct difference. (b) Name the mistake.

Solution (a): The greatest value with a mark above it is 5; the least is 1. A difference is a subtraction: $5 - 1 = 4$. $\boxed{4}$

Solution (b) — instructor-judged. A full-credit answer says a difference means subtract, and that the student added instead. Model answer: "Difference means how far apart, so you subtract: $5 - 1 = 4$. They added and got 6, which is not a distance between the two numbers." Writing only the number 4 is not full credit.

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6. Shape Y placed in the "two pairs of parallel sides" group. (a) How many pairs does Y have? (b) Name the mistake.

Solution (a): Shape Y is a trapezoid. Its top and bottom sides stay the same distance apart — that is one pair. Its two slanted sides lean towards each other and would meet if extended, so they are not a pair. $\boxed{1}$

Solution (b) — instructor-judged. A full-credit answer says having four sides does not guarantee two pairs of parallel sides, and points at the slanted sides. Model answer: "Only the top and bottom are parallel. The two slanted sides get closer together, so they are not parallel. That is one pair, so it does not belong in that group." Naming the shape without saying which sides are parallel is not full credit.

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7. A scale from 1 to 5 with one gap twice as wide. (a) How many equal gaps should there be? (b) Why do uneven gaps mislead?

Solution (a): From 1 to 5 counting by 1 there are five ticks, and the gaps sit *between* the ticks: $(5-1)\div 1 = 4$. Counting ticks instead of gaps gives 5 and is the usual slip here. $\boxed{4}$

Solution (b) — instructor-judged. A full-credit answer explains the visual consequence: with a wider gap, the same number of marks spreads over more room, so a shorter column can look like the biggest one and the plot gives a false picture at a glance. Saying only that it looks untidy or uneven is not full credit.

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8. A copy of Line Plot P has only 11 marks instead of 12. (a) How many are missing? (b) How can you find which value was left out?

Solution (a): $12 - 11 = 1$. $\boxed{1}$

Solution (b) — instructor-judged. A full-credit answer gives a method that does not restart the plot: compare the two plots column by column and find the single column that is one mark short. Model answer: "I would check each column against the real plot — 2, 3, 4, 2, 1. The one column that has one fewer mark is the value that was left out." Saying only "count them all again" is not full credit, because it is not a diagnosis.